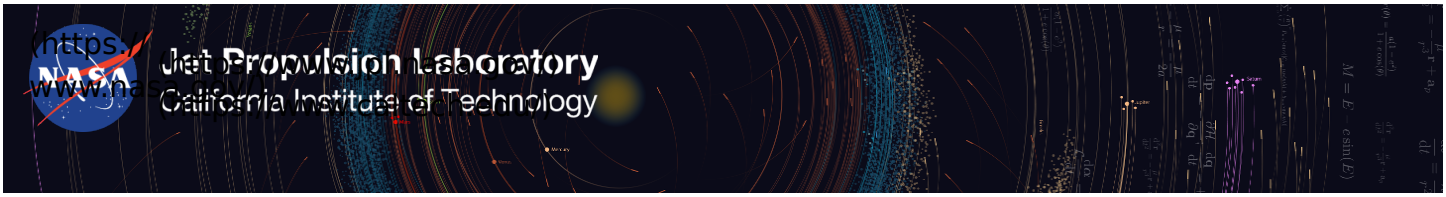

Small-Body Element Tables

Orbital elements below are available as fixed-format ASCII tables. Asteroid elements are available as Keplerian elements while comet elements use perihelion distance and time of perihelion passage instead of semimajor axis and mean anomaly. See the table column headings description below. Magnitude parameters, H and G, are included in the asteroid tables (for a few unnumbered asteroids, H and G are unknown and represented as `99.00` and `0.00`, respectively). If you need other parameters, you should use the Small-Body Database Query (/tools/sbdb_query.html) tool.

The files below were updated on 2026-04-10 09:33:32 UTC.

- 887103 Numbered Asteroids (/dat/ELEMENTS.NUMBR) (96.4 MB) or gzip compressed file (/dat/ELEMENTS.NUMBR.gz) (36.2 MB)
- 633041 Unnumbered Asteroids (/dat/ELEMENTS.UNNUM) (61.7 MB) or gzip compressed file (/dat/ELEMENTS.UNNUM.gz) (22.8 MB)
- 4061 Comets (/dat/ELEMENTS.COMET) (507.4 KB)

Sample Numbered Asteroid File Header

```
 Num   Name           Epoch     a         e        i        w        Node      M        H    G   Ref
------ ----------------- ----- ---------- ---------- --------- --------- --------- ----------- ----- ----- ----------
     1 Ceres          55400  2.7653485 0.07913825  10.58682  72.58981  80.39320 113.4104434  3.
     2 Pallas         55400  2.7721532 0.23099956  34.84090 310.15094 173.12950  96.1482660  4.
     3 Juno           55400  2.6700912 0.25498122  12.98211 248.10807 169.91138  32.0960832  5.
 ...
```

Sample Unnumbered Asteroid File Header

```
Designation   Epoch     a          e         i         w         Node       M          H    G   Ref
------------- ----- ----------- ---------- --------- --------- --------- ----------- ----- ---- ----------
  1927 LA     25051   3.3440715 0.33361827  17.63150 341.10951 191.71742  45.7201727 11.00
  1935 UZ     28097   2.1575113 0.25129007   4.75775 281.18859 134.51847 342.3961099 99.00
  ...
A/2018 V3     58643 121.4442887 0.98897009 164.97713   3.49875 308.65536 359.9324692 15.8
  ...
```

Sample Comet File Header

```
Num  Name                                        Epoch     q          e         i         w         Node        Tp       Ref
---------------------------------------------- ------- ----------- ---------- --------- --------- --------- --------------- ------------
  1P/Halley                                    49400  0.58597811 0.96714291 162.26269 111.33249  58.42008 1
  2P/Encke                                     55460  0.33586872 0.84833156  11.78308 186.54970 334.56582 2
  3D/Biela                                     -9480  0.87907300 0.75129900  13.21640 221.65880 250.66900 18
  ...
```

ASCII File Column Headings

column	units	Description
Num		Number assigned to the asteroid or comet. Numbers are only assigned to secure short-period (<200 year) comets.
Name		Name given to the numbered asteroid or comet.
Designation		Unnumbered asteroid's provisional designation.
Epoch	(TDB)	Epoch of the elements represented as the Modified Julian Date (MJD), which is defined as the Julian date - 2400000.5.
a	(au)	Semimajor axis (/glossary/semimajor_axis.html) of the orbit (asteroids only).
q	(au)	Perihelion (/glossary/perihelion.html) distance (comets only).
e		Eccentricity of the orbit. (/glossary/eccentricity.html)
i	(deg)	Inclination (/glossary/inclination.html) of the orbit with respect to the J2000 ecliptic plane (/glossary/ecliptic_plane.html).
w	(deg)	Argument of perihelion (/glossary/arg_peri.html) (J2000-Ecliptic).
Node	(deg)	Longitude of the ascending node (/glossary/node.html) (J2000-Ecliptic).
M	(deg)	Mean anomaly (/glossary/ma.html) at epoch (asteroids only).
Tp		Time of perihelion passage (/glossary/tp.html) (comets only), formatted as a calendar date (YYYYMMDD.DDD) where "YYYY" is the year, "MM" is the numeric month, and "DD.DDD" is the day and day fraction.
H	(mag)	Absolute magnitude (/glossary/h.html) (asteroids only and set to "99.00" when unknown).

column	units	Description
G		Magnitude slope parameter (asteroids only and set to "0.00" when H is unknown).
Ref		Orbit solution reference.

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